

PREDICTIVE ANALYSIS OF UNEMPLOYMENT RATES IN INDIA

A Data Science Research Report / Whitepaper

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Research Topic

Predictive Analysis of Unemployment Rates in India Using Machine Learning

GitHub Repository

Unemployment Analysis – Reproducible Research & Predictive Modeling

ABSTRACT

Unemployment is an important economic and social issue that affects individuals, families, businesses, and government policy. The COVID-19 pandemic caused major disruptions in employment and labour markets across India. This research analyzes unemployment trends across Indian states and regions and develops predictive models to study whether historical labour-market variables can help estimate unemployment rates.

The study uses the Unemployment in India dataset containing information on region, date, area, estimated unemployment rate, estimated employed population, and estimated labour participation rate. After data cleaning, 740 valid observations were used for descriptive analysis. The average unemployment rate increased from 9.51% during the Pre-COVID period to 17.77% during the COVID period.

For predictive modeling, historical unemployment rates were incorporated through one-month, two-month, and three-month lag features. Linear Regression and Random Forest Regression models were trained using an 80:20 train-test split. The Random Forest model achieved a Mean Absolute Error (MAE) of 4.954, while Linear Regression achieved an MAE of 5.240. Linear Regression produced a slightly better RMSE and R^2 score on the final test set.

The results indicate that employment level, labour participation rate, and historical unemployment values provide useful information for unemployment analysis. However, the negative R^2 values indicate that the models have limited generalization performance on the final test set. Therefore, the study presents the predictive results as exploratory rather than as a highly accurate forecasting system.

Keywords: Unemployment, COVID-19, Data Science, Machine Learning, Linear Regression, Random Forest, Predictive Modeling, India

1. INTRODUCTION

Unemployment is one of the important indicators used to understand the condition of an economy and its labour market. A high unemployment rate can affect household income, living standards, business activity, and economic growth. Therefore, analyzing unemployment trends can help researchers and policymakers understand changes in employment conditions.

The COVID-19 pandemic created significant disruptions in employment and economic activity in India. During this period, many businesses were affected, economic activities were restricted, and employment opportunities changed considerably. The unemployment rate therefore showed a noticeable increase during the COVID-19 period compared with the period before the pandemic.

Data Science and Machine Learning provide methods for analyzing large datasets and identifying patterns in historical data. In this research, Python-based data analysis techniques are used to study unemployment trends across Indian states and regions. Descriptive analysis is first performed to compare unemployment before and during the COVID-19 period. Machine Learning models are then developed using employment, labour participation, and historical unemployment values.

Two predictive models are evaluated in this study: Linear Regression and Random Forest Regression. Their performance is compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

The main purpose of this research is not to claim a perfect unemployment forecasting system, but to investigate whether historical labour-market information can provide useful predictive information and to understand the limitations of the selected models.

2. RESEARCH QUESTION AND OBJECTIVES

2.1 Research Question

The main research question of this study is:

How did the unemployment rate change across Indian states and regions during the COVID-19 period compared with the period before COVID-19, and can historical labour-market variables be used to predict unemployment rates?

2.2 Objectives

The main objectives of this research are:

1. To analyze unemployment trends across different Indian states and regions.
2. To compare the average unemployment rate between the Pre-COVID and COVID periods.
3. To examine the relationship between unemployment, employment, and labour participation.
4. To create historical lag features from previous unemployment-rate observations.
5. To develop predictive models using Linear Regression and Random Forest Regression.
6. To evaluate the predictive models using MAE, RMSE, and R^2 metrics.
7. To identify the important variables contributing to the Random Forest predictions.
8. To discuss the practical implications and limitations of the predictive analysis.

2.3 Research Hypotheses

H1: The average unemployment rate was higher during the COVID-19 period than during the Pre-COVID period.

H2: Historical unemployment rates and labour-market variables contain useful information for predicting unemployment rates.

H0: There is no meaningful difference in unemployment levels between the Pre-COVID and COVID periods

3. LITERATURE BACKGROUND

Previous research has shown that the COVID-19 pandemic had a significant impact on employment and labour markets in India. Researchers have used unemployment and labour-market datasets to study changes in employment conditions and to understand the effect of the pandemic on workers and economic activity.

Machine Learning has also been applied to unemployment-rate prediction. Regression-based methods can be used to identify relationships between unemployment and other labour-market variables, while ensemble methods such as Random Forest can capture more complex relationships in data.

Studies on unemployment forecasting have demonstrated the potential of supervised Machine Learning models for predicting unemployment rates. These studies suggest that historical unemployment observations and related economic variables can provide useful information for predictive analysis.

However, unemployment is influenced by many factors, including economic conditions, regional differences, seasonal effects, government policies, migration, and unexpected events. Therefore, a model based only on a limited set of historical labour-market variables may have limited predictive accuracy.

This research builds on these ideas by combining descriptive COVID-period analysis with a predictive modeling workflow. Linear Regression and Random Forest Regression are used to evaluate whether historical unemployment rates, estimated employment, and labour participation can provide useful predictive information.

4. DATA AND METHODOLOGY

4.1 Dataset Description

This study uses the “Unemployment in India” dataset for analyzing unemployment trends and developing predictive models. The dataset contains observations from different Indian states and regions over time.

The main variables in the dataset are:

- **Region:** Indian state or region.
- **Date:** Date of the observation.
- **Frequency:** Frequency of the recorded observation.
- **Estimated Unemployment Rate (%):** Percentage of unemployed people in the labour force.
- **Estimated Employed:** Estimated number of employed people.
- **Estimated Labour Participation Rate (%):** Percentage of the working-age population participating in the labour force.
- **Area:** Classification of the observation as Rural or Urban.

The original dataset contained 768 rows and 7 columns. During data cleaning, 28 rows containing missing values were removed. The final cleaned dataset contained 740 valid observations.

4.2 Data Cleaning

The following preprocessing steps were performed:

1. Column names were cleaned by removing unnecessary spaces.
2. Missing observations were identified and removed.
3. The Date column was converted into a proper date format.
4. Extra spaces were removed from categorical columns.
5. The observations were sorted chronologically.
6. The cleaned dataset was used for descriptive analysis and predictive modeling.

4.3 COVID-19 Period Classification

To study the effect of the pandemic period, observations were divided into two groups:

- **Pre-COVID:** Dates before March 2020.
- **COVID:** Dates from March 2020 onward.

4.4 Feature Engineering

Three lag variables were created:

- **Lag_1:** Previous unemployment-rate observation.
- **Lag_2:** Unemployment rate from two previous observations.
- **Lag_3:** Unemployment rate from three previous observations.

4.5 Predictive Modeling

Five variables were used as predictors:

1. Estimated Employed
2. Estimated Labour Participation Rate (%)
3. Lag_1
4. Lag_2
5. Lag_3

4.6 Train-Test Split

The modeling dataset was divided into training and testing data using an 80:20 split.

- Training observations: 524
- Testing observations: 51

The training data was used to fit the models, while the final test data was used to evaluate their predictive performance.

4.7 Evaluation Metrics

Three evaluation metrics were used:

Mean Absolute Error (MAE)

Root Mean Squared Error (RMSE)

R² Score

5. EXPLORATORY DATA ANALYSIS AND RESULTS

5.1 Overall Unemployment Trend

Exploratory Data Analysis (EDA) was performed to understand the distribution and trends in unemployment rates across Indian states and regions.

The analysis considered the unemployment rate as the main outcome variable and examined its variation across time, regions, and the COVID-19 period.

The dataset showed considerable variation in unemployment rates among different regions. This indicates that unemployment conditions were not uniform across India.

5.2 Pre-COVID and COVID Comparison

The unemployment rate was compared between the Pre-COVID and COVID periods.

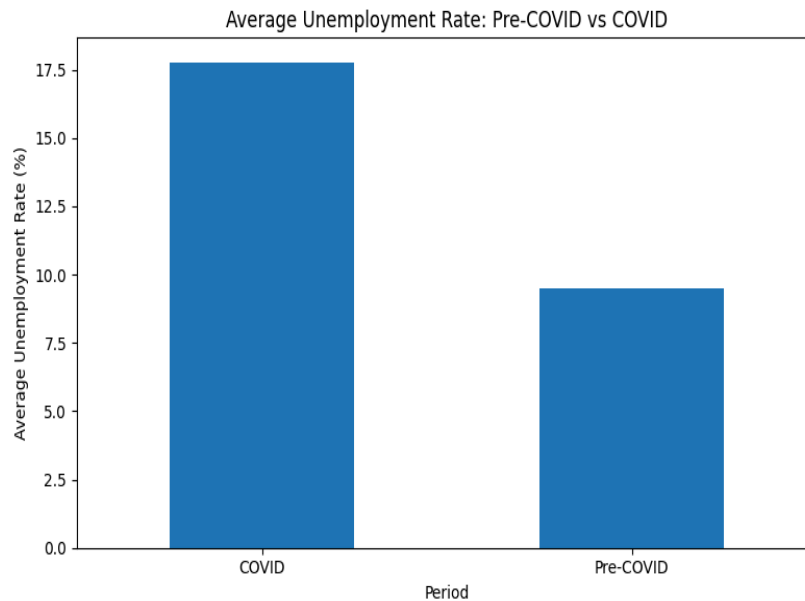
The results were:

Period.	 Average Unemployment Rate
Pre-COVID	9.51%
COVID.	17.77%

The average unemployment rate increased substantially during the COVID period. The difference between the two averages was approximately 8.26 percentage points.

This result supports the research hypothesis that unemployment was higher during the COVID period than during the Pre-COVID period.

Figure 1: Average Unemployment Rate – Pre-COVID vs COVID



The corresponding graph from the analysis notebook visually shows the difference between the two periods.

5.3 Unemployment Rate by State/Region

The average unemployment rate was also calculated for individual states and regions.

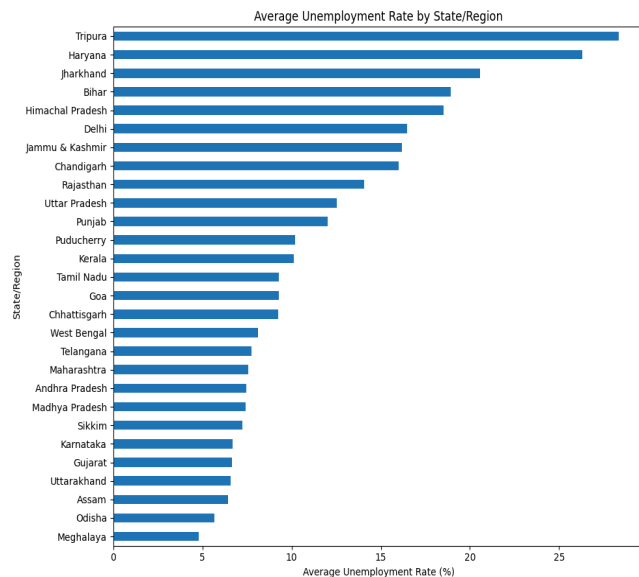
Some regions with relatively higher average unemployment rates included:

- Tripura – 28.35%
- Haryana – 26.28%
- Jharkhand – 20.59%
- Bihar – 18.92%
- Himachal Pradesh – 18.54%

Some regions with comparatively lower average unemployment rates included:

- Meghalaya – 4.80%.
- Assam – 6.43%
- Odisha – 5.66%.
- Karnataka – 6.68%.
- Uttarakhand – 6.5

Figure 2: Average Unemployment Rate by State/Region



The state/region comparison graph provides a visual representation of these differences.

5.4 Interpretation of EDA Findings

The descriptive analysis provides three important observations.

First, the average unemployment rate was considerably higher during the COVID period than during the Pre-COVID period.

Second, unemployment varied substantially across Indian states and regions. Some regions recorded much higher average unemployment than others.

Third, these regional and temporal differences suggest that unemployment is influenced by multiple factors and that a single national-level pattern may not fully represent local labour-market conditions.

6. PREDICTIVE MODELING

6.1 Model Development

After completing the exploratory analysis, predictive models were developed to estimate the unemployment rate using labour-market variables and historical unemployment observations.

The target variable was Estimated Unemployment Rate (%). Five predictor variables were used:

- Estimated Employed
- Estimated Labour Participation Rate (%)
- Lag_1
- Lag_2
- Lag_3

The lag variables represent previous unemployment-rate observations for the same region and area. This approach allows the models to use historical information when making predictions.

Two supervised Machine Learning models were evaluated:

1. Linear Regression
2. Random Forest Regression

6.2 Linear Regression Results

Linear Regression was used as the baseline model.

The model achieved the following results on the test dataset:

- **MAE:** 5.240
- **RMSE:** 6.198
- **R² Score:** -0.194

The results indicate that the average absolute prediction error was approximately 5.24 percentage points. The negative R^2 score indicates that the model did not outperform a simple mean-based baseline on the final test data.

6.3 Random Forest Results

A Random Forest Regression model was also trained using the same predictor variables.

The model achieved:

- **MAE:** 4.954
- **RMSE:** 6.220
- **R^2 Score:** -0.203

Random Forest produced a slightly lower MAE than Linear Regression, indicating a small improvement in average absolute error. However, its RMSE and R^2 were slightly worse than those of Linear Regression.

6.4 Model Comparison

Model.	MAE	RMSE.	R^2 Score
Linear Regression.	5.240.	6.198.	-0.194
Random Forest.	4.954.	6.220.	-0.203

The comparison shows that neither model can be considered an overall superior model based on all three evaluation metrics.

Random Forest achieved the lowest MAE, while Linear Regression achieved slightly better RMSE and R^2 values.

Therefore, the predictive performance of both models should be considered limited on the final test set.

7. FEATURE IMPORTANCE AND INTERPRETATION

7.1 Random Forest Feature Importance

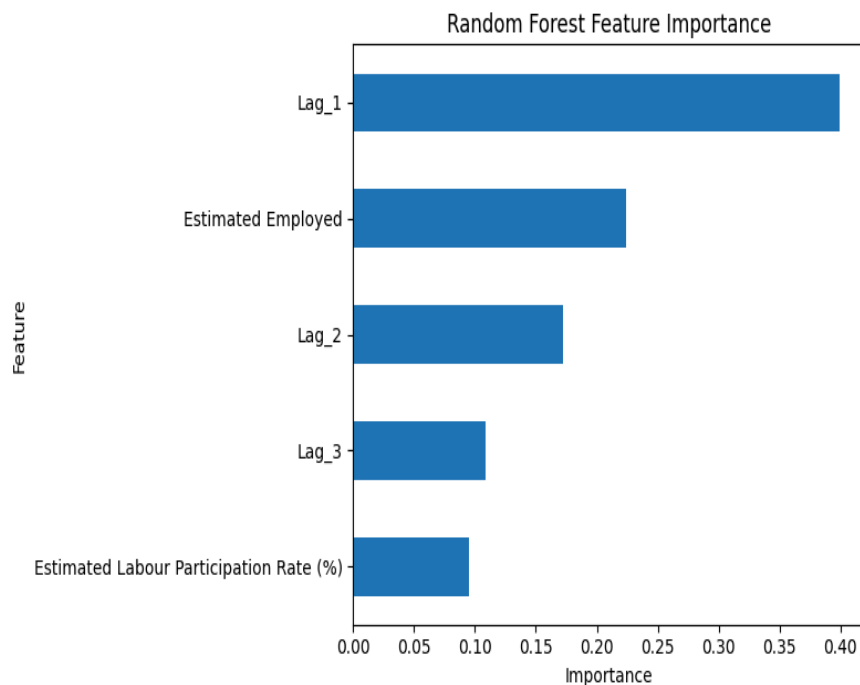
The Random Forest model was used to identify the relative importance of the variables used for prediction. Feature importance indicates how much each input variable contributed to the predictions made by the model.

The feature importance values were:

- Estimated Employed: 25.75%
- Estimated Labour Participation Rate (%): 22.95%
- Lag_1: 18.56%
- Lag_3: 17.19%
- Lag_2: 15.53%

Estimated Employed was the most important feature, followed by the Estimated Labour Participation Rate. The historical unemployment features also contributed to the model, showing that previous unemployment observations contain useful information for prediction.

Figure 3: Random Forest Feature Importance

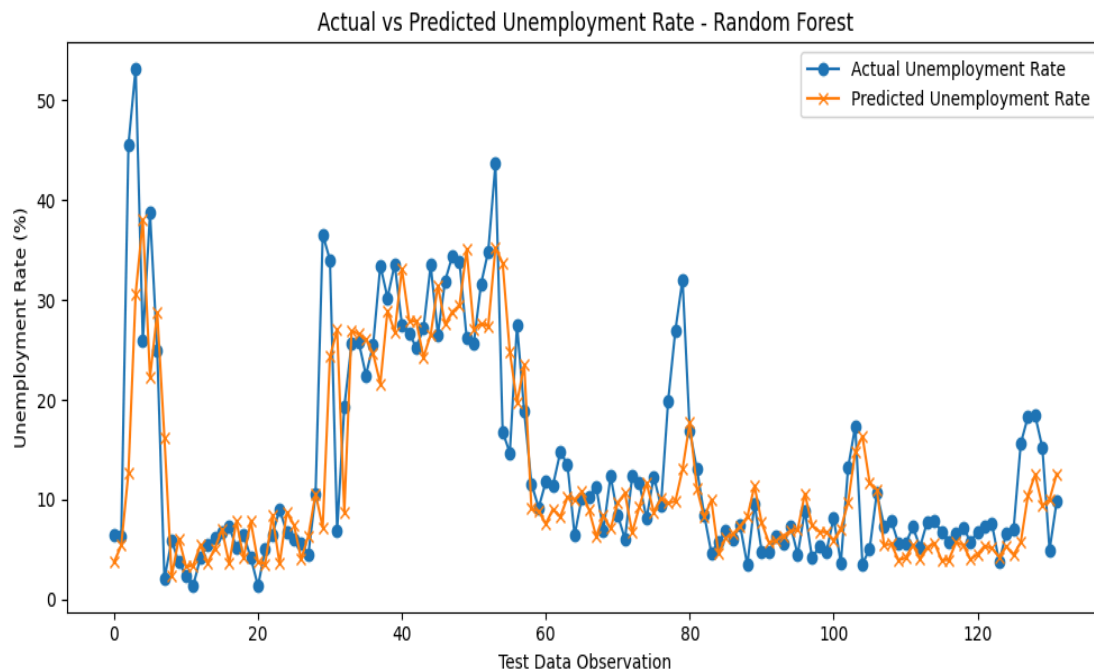


The feature-importance graph from the analysis notebook presents the relative contribution of each predictor.

7.2 Actual vs Predicted Unemployment Rate

The predicted unemployment rates from the Random Forest model were compared with the actual unemployment rates in the test dataset. The comparison helps to visually evaluate how closely the model predictions follow the observed values.

Figure 4: Actual vs Predicted Unemployment Rate – Random Forest



The graph shows differences between actual and predicted values across the test observations. These differences are consistent with the evaluation results, which indicate that the models have limited predictive accuracy on the final test set.

8. BUSINESS AND POLICY IMPLICATIONS

8.1 Business Implications

Unemployment trends provide useful information for businesses, organizations, and workforce planners. Changes in unemployment can influence consumer spending, labour availability, and demand for products and services.

The results of this study show that employment levels and labour participation rates are important variables in the predictive model. Businesses can use labour-market information to understand changes in workforce availability and to support recruitment and workforce planning decisions.

8.2 Policy Implications

The significant increase in unemployment during the COVID-19 period highlights the importance of monitoring labour-market conditions during economic disruptions.

Government agencies and policymakers can use unemployment data to identify regions experiencing higher unemployment and to design targeted employment-support programs.

Data-driven monitoring can also support decisions related to skill-development programs, employment generation, worker support, and regional economic planning.

8.3 Use of Predictive Analytics

Predictive analytics can help organizations monitor historical patterns and identify possible changes in unemployment conditions. However, accurate unemployment forecasting requires additional information such as economic growth, inflation, migration, seasonal patterns, industry-level employment, government policies, and other macroeconomic indicators.

9. LIMITATIONS

This study has several limitations that should be considered while interpreting the results.

9.1 Dataset Limitations

The analysis is based on a single unemployment dataset containing information on selected labour-market variables. Other important factors such as inflation, GDP growth, migration, industry-level employment, education, government policies, and seasonal effects were not included.

9.2 Limited Predictive Variables

The predictive models use estimated employment, labour participation rate, and historical unemployment values. Additional economic and social variables could potentially improve predictive performance.

9.3 Model Performance

Both Linear Regression and Random Forest produced negative R^2 values on the final test set. This indicates that the models did not outperform a simple constant-mean baseline for the test observations. Therefore, the models should not be considered highly accurate forecasting systems.

9.4 Regional Differences

Unemployment conditions can vary considerably across Indian states and between rural and urban areas. A model trained using the available variables may not fully capture these regional differences.

9.5 COVID-19 as an Exceptional Event

The COVID-19 period involved unusual economic and employment conditions. Relationships observed during this period may not necessarily represent normal economic conditions.

9.6 Future Improvements

Future research could include larger and more recent datasets, additional economic indicators, seasonal features, advanced time-series models, and more robust time-based validation techniques. Models such as SARIMA, Prophet, XGBoost, or other advanced forecasting approaches could also be investigated.

10. CONCLUSION

This study analyzed unemployment trends across Indian states and regions and investigated the use of Machine Learning for predictive unemployment analysis.

The exploratory analysis showed a clear increase in the average unemployment rate during the COVID-19 period. The average unemployment rate increased from 9.51% during the Pre-COVID period to 17.77% during the COVID period, representing an increase of approximately 8.26 percentage points.

Two predictive models, Linear Regression and Random Forest Regression, were developed using estimated employment, labour participation, and historical unemployment values. Random Forest achieved a slightly lower MAE of 4.954, while Linear Regression achieved a slightly better RMSE of 6.198 and R^2 of -0.194.

The negative R^2 values indicate that the selected models had limited predictive performance on the final test set. Therefore, the results should be interpreted as an exploratory predictive analysis rather than a highly accurate unemployment forecasting system.

The feature-importance analysis showed that Estimated Employed and Estimated Labour Participation Rate were the two most important variables in the Random Forest model. Historical unemployment values also contributed to the predictions.

Overall, this research demonstrates a complete and reproducible Data Science workflow involving data cleaning, exploratory analysis, feature engineering, predictive modeling, model evaluation, and interpretation. The study also highlights the importance of using richer datasets and additional economic variables for future unemployment forecasting research.

11. REPRODUCIBILITY AND REPOSITORY STRUCTURE

This research follows a reproducible Data Science workflow. The analysis notebook, research documentation, dataset structure, dependencies, and testing files are maintained in a GitHub repository.

11.1 GitHub Repository

Repository: Unemployment Analysis – Reproducible Research & Predictive Modeling

GitHub Link:

<https://github.com/rukminichakurr-lgtm/unemployment-analysis-reproducible>

11.2 Repository Structure

The repository contains the following main components:

- **raw/** – Contains the original dataset used for the analysis.
- **interim/** – Contains intermediate data generated during the workflow.
- **processed/** – Contains processed data used for analysis.
- **notebooks/** – Contains the Jupyter Notebook with the complete analysis and predictive modeling workflow.
- **report/** – Contains research report documentation.
- **tests/** – Contains synthetic tests used to check parts of the analysis pipeline.
- **README.md** – Provides an overview of the project and instructions.
- **preregistration.md** – Documents the research question, hypotheses, variables, exclusions, and planned analysis.
- **requirements.txt** – Lists the Python dependencies required for the project.
- **requirements-lock.txt** – Provides fixed package versions for reproducibility.

12. REFERENCES

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